

$$\dot{n}_{\alpha k}(t) = \sum_p \Gamma_{p \rightarrow \alpha k} n_{\alpha p} (1 - n_{\alpha k}) - \Gamma_{\alpha k \rightarrow p} (1 - n_{\alpha p}) n_{\alpha k} =$$

$$= A (1 - n_{\alpha k}) - B n_{\alpha k}$$

logit transformation:

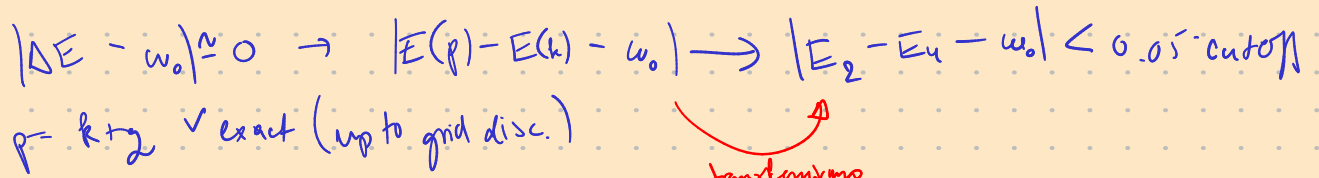
$$n_{\alpha k}(\theta) = \frac{1}{1 + e^{\gamma_{\alpha k}(\theta)}} \quad \left| \frac{d}{dt} \right.$$

$$\dot{n} = \frac{e^{\gamma}}{(1+e^{\gamma})^2} \cdot \dot{\gamma} = \left[ \frac{1}{(1+e^{\gamma})} \right] \left( \frac{1}{1+e^{-\gamma}} \right) \dot{\gamma} =$$

$$\dot{\gamma}_{\alpha k} = \frac{\dot{n}_{\alpha k}}{(1 - n_{\alpha k}) n_{\alpha k}} = \frac{A(1 - n_{\alpha k}) - B n_{\alpha k}}{(1 - n_{\alpha k}) n_{\alpha k}} =$$

$$= \frac{A}{n_{\alpha k}} - \frac{B}{1 - n_{\alpha k}}$$

$$\Rightarrow \boxed{\dot{\gamma}_i = \frac{\sum_j \Gamma_{j \rightarrow i} n_j}{n_i} - \frac{\sum_j \Gamma_{i \rightarrow j} (1 - n_j)}{1 - n_i}}$$



$$y_0 = -iB \frac{1}{z^2 + k^2} \cdot \frac{2}{\sqrt{\omega}}$$

$$|g_{ad}|^2 = \cancel{|-i|^2} \left| \frac{z^2}{(z^2 + z_c^2)} \frac{1}{w} \right| \left[ A u_{k+z} u_k + B v_{k+z} v_k \right]^2$$

→ prod bathes:  $\dot{n}_k = \sum_p \Gamma_{k \leftarrow p} n_p (1 - n_k) - \sum_p \Gamma_{p \leftarrow k} (1 - n_p) n_k$

$$\frac{\delta_{ij}(T_1)}{\delta_{ij}(T_2)} = \frac{f\left(\frac{E_i}{T_1}\right) f\left(\frac{-E_j}{T_1}\right)}{f\left(\frac{E_i}{T_2}\right) f\left(\frac{E_j}{T_2}\right)} = \frac{(1+e^z)(1+e^w)e^y}{(1+e^x)(1+e^y)e^w} = e^{y-w} \left[ \frac{f(E_i/T_1) f(E_j/T_1)}{f(E_i/T_2) f(E_j/T_2)} \right]$$

$$\frac{(1+e^{z+w}+e^z+e^w)}{(1+e^{x+y}+e^x+e^y)} e^{y-w} =$$

$$= \frac{1+e^{z+w}+e^z+e^w}{1+1+1+1} e^{y-w} = \frac{1}{4} \left[ \underbrace{e^{-w} + e^z}_{2 \operatorname{sh}(x)} + \underbrace{e^{z-w} + 1}_1 \right] =$$

$$= \frac{1}{2} \left( 1 + \operatorname{sh}\left(\frac{E}{T \rightarrow 0}\right) \right)$$

$$\begin{array}{l} T_1 \rightarrow \infty : x \rightarrow 0, y \rightarrow 0 \\ T_2 \rightarrow 0 : z \rightarrow \infty, w \rightarrow \infty \end{array}$$

$$\frac{\Gamma_{ij}}{\Gamma_{ji}} = \frac{\delta_{ij}(T_1) + \delta_{ij}(T_2)}{\delta_{ji}(T_1) + \delta_{ji}(T_2)} = ?$$

$$\frac{\Gamma_{ij}}{\Gamma_{ji}} = \frac{g_{ij}}{g_{ji}} \frac{([1+N(w)]\delta(E_j-E_i+w) + N(w)\delta(E_j-E_i-w))}{([1+N(w)]\delta(E_i-E_j+w) + N(w)\delta(E_i-E_j-w))}$$

$$\frac{1}{e^x - 1} = \frac{e^{-x}}{1 - e^{-x}} = \frac{1}{e^x - 1}$$

$$\mathcal{D} = \mathcal{D}(T_1) + \mathcal{D}(T_2)$$

→ bei bath 1 = bath 2 haben je Temperatur:

$$E = E_{pL} - E_{vp} \quad , \quad A(\epsilon) \approx \delta(\epsilon - \omega) \\ A(-\epsilon) \approx \delta(-\epsilon - \omega) = \delta(\epsilon + \omega)$$

$$\Gamma_{\text{hy}}^{\text{tr}} = 2\pi |g|^2 \cdot \underbrace{A(|\epsilon|)}_{\text{reduziert auf } T} \left\{ \left( N(|\epsilon|, T_1) + \delta(-\epsilon) \right) + \left( N(|\epsilon|, T_2) + \delta(-\epsilon) \right) \right\} =$$

$$\epsilon > 0 : A(\epsilon) N(\epsilon) \quad \checkmark$$

$$\epsilon < 0 : A(-\epsilon) (1 + N(\epsilon)) \quad \checkmark$$

$$\Gamma_{\text{hy}}^{\text{tr}} = 2\pi |g|^2 A(|\epsilon|) \cdot \left\{ 2\delta(-\epsilon) + N(|\epsilon|, T_1) + N(|\epsilon|, T_2) \right\}$$

$$\epsilon > 0 : 2 \cdot \left\{ 1 + \frac{N_1 + N_2}{2} \right\} = 2(1 + \bar{N}) \quad \text{dividiert man ein } \bar{N} \text{ heraus}$$

$$\epsilon < 0 : 2 \cdot \left\{ \frac{N_1 + N_2}{2} \right\} = 2\bar{N}$$

$$\rightarrow \frac{\Gamma(\epsilon)}{\Gamma(-\epsilon)} = \frac{1 + \bar{N}}{\bar{N}} = e^{\epsilon/T^*} \rightarrow$$

$$T^* \triangleq \frac{\omega_0}{\ln\left(1 + \frac{1}{\bar{N}(\omega_0)}\right)} \quad , \quad A \sim \delta$$

radiative hopping:

$$\Gamma_{\text{hy}}^{\text{tr}} = 2\pi |g|^2 \left\{ \left( \delta(-\epsilon) + N(|\epsilon|, T_1) \right) \cdot \delta(|\epsilon| - \omega_1) + \left( \delta(-\epsilon) + N(|\epsilon|, T_2) \right) \delta(|\epsilon| - \omega_2) \right\}$$

$$\frac{\Gamma(\epsilon)}{\Gamma(-\epsilon)} = \frac{(1 + N_1) \delta(\epsilon - \omega_1) + (1 + N_2) \delta(\epsilon - \omega_2)}{N_1 \delta(\epsilon + \omega_1) + N_2 \delta(\epsilon + \omega_2)} = e^{\epsilon/T^*}$$

case 1:  $\Gamma_{\omega_0}^{\mu} = 2\pi |g|^2 A(|\epsilon|) (2\theta(-\epsilon) + N_1 + N_2)$

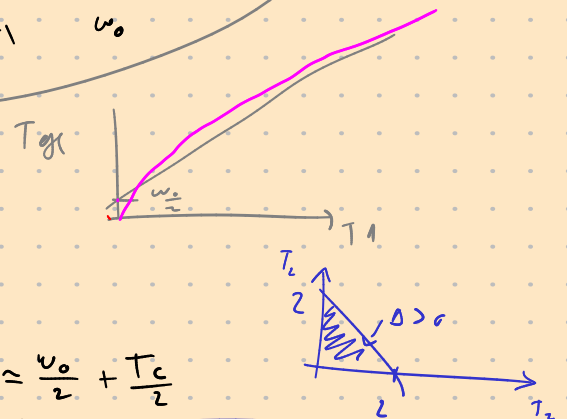
$\Gamma_{\omega}^{\mu} = 2\pi |g|^2 A(|\epsilon|) (2\theta(-\epsilon) + \bar{N}(T_{\text{eff}}))$   $T_{\text{eff}}/T_c = 1 \rightarrow \Delta = 0$  ?

$T_{\text{eff}} \approx \frac{\omega_0}{\ln(1 + \frac{1}{\bar{N}})} \propto \omega_0 \rightarrow$  reino super:  $T_2 = 0 : N_2 = N(\epsilon, 0) \approx 0$   
 $\rightarrow N_1 > 0, \bar{N} = \frac{N_1}{2}$

$\rightarrow T_{\text{eff}} \approx \frac{\omega_0}{\ln(1 + \frac{2}{N(\omega_0, T_1)})} = T_{\text{eff}}(T_1) \approx \frac{\omega_0}{\ln(1 + \frac{2\omega_0}{T_1})}$

$\rightarrow T_1 \gg T_c \rightarrow N(\omega_0, T_1) = \frac{1}{e^{\frac{\omega_0}{T_1}} - 1} \approx \frac{1}{\frac{\omega_0}{T_1} + 1 - 1} = \frac{T_1}{\omega_0}$   
 $\ll 1$

$\approx \omega_0 \cdot \left( \frac{T_1}{2\omega_0} + \frac{1}{2} \right) \approx \boxed{\frac{\omega_0}{2} + \frac{T_1}{2}}$



$\rightarrow T_{\text{eff}} \approx \frac{\omega_0}{2} + \frac{T_1}{2} \rightarrow \text{for } T_1 = T_c : T_{\text{eff}} \approx \frac{\omega_0}{2} + \frac{T_c}{2}$

$\hookrightarrow$  defining  
 $T_c = 2T_{\text{eff}} - \omega_0$

$\rightarrow$  best way:  $T_c \approx \frac{2\omega_0}{e^{\frac{\omega_0}{T_{\text{eff}}}} - 1} = 2\omega_0 N(\omega_0, T_{\text{eff}})$

$\rightarrow$  analitika  $\Delta = 0$  formula:

$T_{\text{eff}}^c \Leftrightarrow T_{\text{eff}} = 1$

$T_2(T_1) = ?$

$\rightarrow 1 = \frac{\omega_0}{\ln(1 + \frac{1}{\bar{N}})} \rightarrow N_1 + N_2 = \frac{2}{e^{\frac{\omega_0}{1}} - 1} = \frac{1}{\underbrace{e^{\frac{\omega_0}{T_1}} - 1}_{N_1}} + \frac{1}{e^{\frac{\omega_0}{T_2}} - 1}$   
 $\propto$

$\frac{1}{\ln(1 + (\alpha - N_1)^{-1})} = T_2/\omega_0 \rightarrow$

$T_2(T_1) = \frac{\omega_0}{\ln\left(1 + \left[\frac{2}{e^{\frac{\omega_0}{1}} - 1} - \frac{1}{e^{\frac{\omega_0}{T_1}} - 1}\right]^{-1}\right)}$

$\left( \frac{2e^{\frac{\omega_0}{T_1}} - 2 - e^{\frac{\omega_0}{1}} + 1}{(e^{\frac{\omega_0}{1}} - 1)(e^{\frac{\omega_0}{T_1}} - 1)} \right)^{-1} = \frac{e^{\omega_0(1 + \frac{1}{T_1})} - e^{\omega_0} - e^{\omega_0/T_1} + 1}{2e^{\omega_0/T_1} - e^{\omega_0} - 1}$

maximalen gap, erreichbar bei  $T_1 = 0$  (d.h.  $T_2 = 0$ , stationär):

$$T_2(u) = T_{\text{eff}}^c(u) = \frac{w_0}{\ln\left(1 + \frac{e^{w_0} - 1}{2}\right)} = \boxed{\frac{w_0}{\ln\left(\frac{e^{w_0} + 1}{2}\right)}}$$